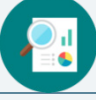
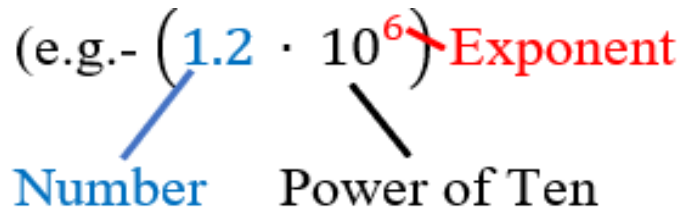


3.9 Writing Small Numbers in Scientific Notation

Lesson Introduction

 Benchmarks	MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency. MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.		 Learning Targets	- I can use the laws of exponents to evaluate numerical expressions with a base of 10 and exponents less than 0. - I can express small numbers in scientific notation. - I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.
 Benchmark Coherence	Building On MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole number exponents and rational number bases.		Working Toward MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational number exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.	
 Essential Questions	- How do you evaluate numerical expressions with bases of 10 and exponents less than zero? - How does place value relate to numerical expressions with bases of 10 and exponents less than zero? - How do you express small numbers in scientific notation? - How many times larger or smaller is the number compared to another number expressed in scientific notation?			
 Lesson Narrative	In this lesson, students apply the laws of exponents to evaluate numerical expressions with bases of 10 and exponents less than zero. Students then explore real-world examples of small numbers and discover how to express them in scientific notation. They also discover how to determine how many times larger or smaller a number is compared to another expressed in scientific notation.			
 Component Summary	3.9.1 Warm-Up (~5 min.)	Square a googol (<i>SE p. 146</i>)	3.9.4 Guided Instruction (10- min.)	Discover how to write small numbers in scientific notation (<i>SE p. 149</i>)
	3.9.2 Exploration (5 min.)	Exploration of powers of 10 (<i>SE p. 147</i>)	3.9.5 Guided Practice (5 min.)	Practice writing small numbers in scientific notation (<i>SE p. 150</i>)
	3.9.3 Bring It Together (10 min.)	Apply laws of exponents to evaluate numerical expressions (<i>SE p. 148</i>)	3.9.6 Collaboration (10 min.)	Compare small numbers written in scientific notation (<i>SE p. 150</i>)
	3.9.7 Wrap-Up (~5 min.)	Match scientific notation to standard or word form (<i>SE p. 151</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Negative powers of 10 <u>Additional Terms:</u> - Scientific notation - Standard form		(Optional) Highlighters or Colored Pencils (Optional) Cards for Wrap-Up	
				Additional Preparation Consider structuring the 3.9.7 Wrap-Up as a card sort. Prepare cards before lesson.

 Teacher Tips	Common Misconceptions • Students may incorrectly think the values of expressions with negative exponents must be negative. • Students may incorrectly think that a number with an exponent of zero equals zero. • Students may believe that they can apply a law of exponents to addition and subtraction expressions. • Students may incorrectly think that the difference in the powers of 10 is how many times larger or smaller a number is than another number. For example, they may think 10^{-5} is two times larger than 10^{-3}, because $(-3) - (-5) = 2$, instead of 10^2 or 100 times larger.	Things to Consider • Encourage students to check their work by expressing the scientific notation in standard form or vice versa. • Consider reviewing place values less than one. • Consider creating an anchor chart with the form of scientific notation as $a \times 10^n$ where $1 \leq a < 10$. Numbers can be expressed using powers of ten that are not in this form, but the definition of scientific notation is that the absolute value of a should be at least 1 and less than 10. Consider using both the mathematical representation of the constraint on a as well as the written description of the constraint. • Consider reviewing the names of place values of the decimal representations of negative powers of 10 or providing them on an anchor chart.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Turn and Talk Prompt students to take turns during 3.9.6 Collaboration. Have students to share their strategies and answers. Then, ask the partner to share if they agree or disagree and why.	MA.K12.MTR.2.1: Multiple Representations In 3.9.4 Guided Instruction students explore and discover how to use multiple representations to express small numbers. Engage students in discussions about how the scientific notation is connected to the standard form of the number.

 Differentiate Instruction	In 3.9.4 Guided Instruction, provide a visual entry point by providing students with a template to use when expressing numbers in scientific notation. In 3.9.6 Collaboration, provide an auditory entry point by prompting students to read their completed statements in questions 1b-1e aloud.	
	Lesson Notes:	

3.9.1 Warm-Up: Googol, not Google

Component Overview: Students are engaged with an interesting application of large numbers, a googol. Students draw on their prior knowledge of the power of a power exponent law to square the number.



3.9.1 Warm-Up: Googol, not Google

A “googol” is a name for a really big number: a 1 followed by 100 zeros.

1. If you square a googol, how many zeros will the answer have? Show your reasoning.

$$(10^{100})^2 = 10^{200}$$

If you square a googol, it will have 200 zeros.



Large numbers could be fascinating for some students and overwhelming to other students. Discuss how expressing the numbers using exponents makes more sense than expressing the numbers in standard form. Prompt students to consider why the popular search engine Google may have chosen a name similar to googol.

3.9.2 Exploration: Powers of Ten

Component Overview: Students use scientific calculators to find the decimal values for powers of 10 less than one. Next, students write the equivalent fractional value and place value for each number. Students discover that the power of 10 is the same as the number of places in the decimal value and the number of zeros in the fractional value. They also discover that the exponent is one less than the total number of places in the place value.



3.9.2 Exploration: Powers of Ten

1. Use a scientific calculator to find the decimal value of each. Then, write the equivalent fractional value.

Power of 10	Decimal Value	Fractional Value
10^0	1.0	$\frac{1}{1}$
10^{-1}	0.1	$\frac{1}{10}$
10^{-2}	0.01	$\frac{1}{100}$
10^{-3}	0.001	$\frac{1}{1,000}$
10^{-4}	0.0001	$\frac{1}{10,000}$
10^{-5}	0.00001	$\frac{1}{100,000}$

2. What do you notice about the relationship between the power of 10 and its value?

The power of ten is the same as the number of places in the decimal value and as the number of zeros in the fractional value.

3. Work with your partner to complete the table.

Power of 10	Fractional Value	Place Value of 1
10^0	1	Ones
10^{-1}	$\frac{1}{10}$	Tenths
10^{-2}	$\frac{1}{100}$	Hundredths
10^{-3}	$\frac{1}{1,000}$	Thousandths
10^{-4}	$\frac{1}{10,000}$	Ten Thousandths
10^{-5}	$\frac{1}{100,000}$	Hundred Thousandths



Consider modeling how to enter negative exponents in calculators for students. Having all students use the same make and model calculator will eliminate potential confusion about how to enter the expressions. For example, you could use the Desmos scientific calculator.



How do you determine the equivalent fractional value? How did you determine the place value of the number?

3.9.2 Exploration: Powers of Ten (continued)

4. Work with your partner to describe how the exponent of the power of ten relates to place value.

The exponent is always one less than the total number of places in the place value.

5. Based on the pattern in the table, what is the value of 10^{-12} , written as a fraction?

$$\frac{1}{1,000,000,000,000}$$



Negative Powers of 10

A base 10 with a negative exponent can be rewritten as a fraction.

$$10^{-a} = \frac{1}{10^a}$$

The value of the fraction relates directly to the place value of the 1 when the quantity is expressed in its equivalent decimal form.



Consider adding this definition to a classroom Word Wall for students to reference. It is not necessary to extend the conversation beyond negative exponents with a base of 10. Students will explore how to evaluate expressions with bases that include rational numbers and algebraic expressions in Unit 4.

3.9.3 Bring It Together: Applying the Laws of Exponents

Component Overview: Students practice applying the Laws of Exponents to write mathematical statements using a power of 10 and then write the decimal equivalent for each simplified expression.



3.9.3 Bring it Together: Applying the Laws of Exponents

The Laws of Exponents can help when rewriting mathematical statements as a power of 10. Sometimes more than one law can be applied.

In the table from the Exploration, $\frac{10^{-11}}{10^{-4}}$ was written as 10^{-7} . Another way to write 10^{-7} is $\frac{1}{10^7}$. Each of these expressions represent the same value.

1. Write each mathematical statement in an equivalent form by applying one or more laws of exponents. Then, write the equivalent decimal value.

Mathematical Statement	Value Written as a Power of 10	Decimal Equivalent
$\left(\frac{1}{10^{-4}}\right)^2$	$\frac{1}{10^{-8}}$	100,000,000.0
$(10^{-11})^3 \times 10^{27}$	10^{-6}	0.000001
$\frac{10^{-13}}{10^{-5}} \cdot 10^8$	10^0	1.0

Numbers like one-billionth, also known as a nano, can be written using powers of 10.

2. Write one-billionth, or $\frac{1}{1,000,000,000}$, as a power of 10.
 10^{-9}
3. One septillionth, 0.000000000000000000000001, is also known as yocto. Write one septillionth as a power of 10.
 10^{-24}



Prompt students to discuss which law(s) of exponents they applied. Ask students to discuss if they agree or disagree and why.

3.9.4 Guided Instruction: Writing Small Numbers in Scientific Notation

Component Overview: Students discover how to express small numbers in scientific notation. Students are presented with real-world examples of small numbers, helping them see the utility of writing small numbers in scientific notation.



3.9.4 Guided Instruction: Writing Small Numbers in Scientific Notation

Powers of ten can also be used to express small numbers in scientific notation.

1. A single red blood cell is about 0.000007 meters in diameter.
 - a. In what place is the 7 in the value, 0.000007?
Millionths
 - b. Which power of 10 relates to this place value?
 10^{-6}
 - c. Express 0.000007 in scientific notation.
 7×10^{-6}
2. Consider the value 0.0000672.
 - a. What value between 1 and 10 could be multiplied by a power of 10 to result in 0.0000672?
6.72
 - b. Which digit is in the ones place of your answer to Part A?
6
 - c. In which place value is that digit in 0.0000672?
hundred thousandths
 - d. Which power of 10 relates to this place value?
 10^{-5}
 - e. Express 0.0000672 in scientific notation.
 6.72×10^{-5}
3. Consider the value 3.4×10^{-9} .
 - a. Complete the statements.
 10^{-9} relates to the *billionths* place. When rewriting 3.4×10^{-9} in standard form, the 3 will be in the billionths place, followed by the 4. The places before the 3 will each hold the digit, 0.
 - b. Express this value in standard form.
0.0000000034



The lesson contains many interesting facts. Engage students in those facts by asking questions such as:

- How do you think they determined that measurement?
- Which do you think is harder to measure – really small things or really large things?



Provide a visual entry point by providing students with a template to use when expressing numbers in scientific notation.



Students continue to explore and discover how to use multiple representations to express small numbers. Engage students in discussions about how scientific notation is connected to the standard form of the number. Rewriting numbers in scientific notation should not be taught by moving decimal places. Rather, in the context of place value and powers of 10 as built-in previous lessons. The Guided Instruction is set up to continue using place value.

3.9.5 Guided Practice: Writing Small Numbers in Scientific Notation

Component Overview: In this Guided Practice, students will use what they learned from the Guided Instruction to determine how to write numbers less than one in scientific notation from a real-world scenario.



3.9.5 Guided Practice: Writing Small Numbers in Scientific Notation

1. The density of oxygen is 1332 millionths of a gram per cubic centimeter.
 - a. Express this value as a decimal in standard form.
0.001332
 - b. Express this value in scientific notation.
 1.332×10^{-3}
2. The Mycoplasmas bacteria is about 2.5×10^{-4} millimeter (mm) in size.
Express this value in standard form.
0.00025



Encourage students to check their work by expressing the scientific notation in standard form or vice versa.



How does writing small numbers in scientific notation compare to writing large numbers in scientific notation? What are the similarities and differences?

3.9.6 Collaboration: Comparing Small Numbers Written in Scientific Notation

Component Overview: Students work with a partner to compare numbers written in scientific notation. Students discover how to determine how many times larger or smaller a number is compared to another expressed in scientific notation.



3.9.6 Collaboration: Comparing Small Numbers Written in Scientific Notation

1. Work with your partner to complete the following.

a. Complete the table.

Scientific Notation	Standard Form
7.2×10^{-4}	0.00072
1.8×10^{-7}	0.00000018

b. Complete the statement to compare the factors being multiplied by powers of 10 in each number that is expressed in scientific notation.

7.2 is 4 times ☐ smaller ☒ larger than 1.8.

c. Complete the statement to compare the powers of 10 in each number that is expressed in scientific notation.

10^{-4} is 1000 times ☐ smaller ☒ larger than 10^{-7} .

d. Complete the statement.

The value 7.2×10^{-4} is 4000 times ☐ smaller ☒ larger than 0.00000018.



Turn and Talk

Prompt students to take turns during this component. Have students share their strategies and answers. Then, ask the partner to share if they agree or disagree and why.



Provide an auditory entry point by prompting students to read their completed statements in questions 1b-1e aloud.

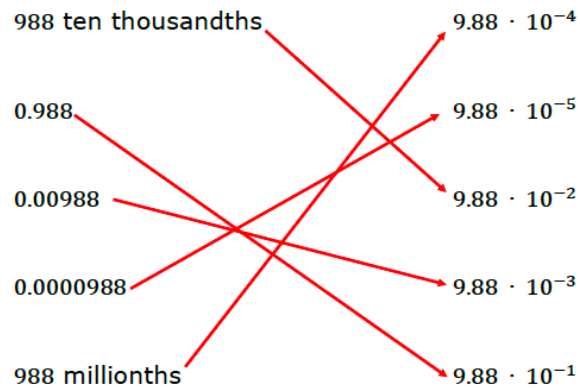
3.9.7 Wrap-Up: Ticket Out the Door

Component Overview: Students match the written and standard forms of numbers with their scientific notation representations. Students should not use calculators. Consider using data from this formative assessment to determine which students need remediation before the next lesson.



3.9.7 Wrap-Up: Ticket Out the Door

- Match each value on the left with its expression in scientific notation on the right by drawing lines to connect the equivalent expressions.



Consider providing a kinesthetic entry point by structuring this component as a card sort.



Challenge students who finish early to research real-world examples of very small numbers and determine how many times larger or smaller a number written in scientific notation is than another number written in scientific notation.